

FLUOR DANIEL

WIND LOAD CALCULATION

- + (Plus sign) means positive pressure acting toward the surface.
- - (Minus sign) means negative pressure acting away from the surface.

Select pressure and force coefficients for main wind-force resisting systems and components and cladding from ASCE 7-95 Figures 6-3 through 6-8 and Tables 6-4 through 6-10. Figure 6-9 provides for full, partial, torsional, and diagonal wind loadings for buildings greater than 60 feet high.

WIND TUNNEL TESTING

ASCE 7-95 permits the use of properly conducted wind tunnel tests for the determination of design wind loads. Refer to ASCE 7-95 Section 6.4.3 for guidance on when such testing is recommended and what elements are necessary for a properly conducted test.

A wind tunnel test conducted in the United States normally costs between \$10,000 to \$50,000. A wind tunnel test cannot be justified unless the expected savings is greater than the cost of the test.

Wind tunnels are commonly booked for use well in advance -- a wind tunnel test should be considered a "long-lead item" and scheduled accordingly.

Boundary Layer Testing

Testing of structures must occur in boundary layer wind tunnels. A boundary layer wind tunnel must have a test section that is sufficiently long to simulate accurately the atmospheric boundary layer from ground to gradient height. Typically, a boundary layer wind tunnel will be longer than 30 feet to allow development of a scale wind pressure that varies with height.

Another common type of wind tunnel is an aeronautical wind tunnel, characterized by uniform air flow and pressure distribution. Testing structures in aeronautical wind tunnels is generally inappropriate.

Additional discussion on boundary layer wind tunnel testing can be found in the reference by Liu.

Contracting Services

Wind tunnel testing services do not lend themselves to typical competitive bid procurement processes. Contracting for wind tunnel testing is similar to contracting for geotechnical services; a desired set of information to support design is indicated, and a detailed scope is recommended by the contractor. Typically, a scope is negotiated with a sole source wind tunnel contractor (consultant). After the scope is mutually agreed upon, commercial terms can be requested and negotiated.

After notice to proceed is issued, the wind tunnel contractor will typically need 2 weeks to construct the model and prepare the wind tunnel. Another 2 weeks is required to obtain the data and prepare a preliminary report.

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VERTICAL VESSELS

Vertical vessels must be designed for along-wind response caused by straight wind (drag forces). Flexible vessels must also consider across-wind response caused by vortex shedding (lift forces). The design procedure herein is also appropriate for determining design wind forces on stacks and chimneys. A vertical vessel (or a stack or chimney) will behave like a cantilever beam. Drag forces will be maximum at the design wind velocity. Lift forces will be maximum at a relatively low wind velocity such as 10 to 30 mph.

Across-wind response procedures are summarized in Attachment 07 and sample design calculations are included in Attachment 01.

General Procedure

The following is derived from ASCE 7-95 Table 6-1 for "Main wind-force resisting systems" of "Open buildings and other structures":

$$F = q_z G C_r A_r$$

where:

F	=	Design wind force distribution on vessel	(pounds)
q_z	=	Velocity pressure, determined as varying with height z	(psf)
G	=	Gust effect factor	(dimensionless)
C_r	=	Force coefficient. Select value from ASCE 7-95 Table 6-7 as described in the Ladders and Piping section following.	(dimensionless)
A_r	=	Projected area of vessel normal to the wind, equal to D times tributary height for each q_z	(ft ²)
D	=	Basic vessel diameter, equal to vessel inside diameter plus 2 times plate (wall) thickness plus 2 times insulation thickness	(ft)

Wind On Appurtenances

The general procedure for vertical vessels requires modification to account for vessel appurtenances such as ladders, piping, and platforms.

Ladders And Piping

Account for ladders and piping only if $D\sqrt{q_z} > 2.5$. In this case, determine C_r as follows:

C_r = The larger of the following:

$$\begin{array}{l} C_{fms} \text{ WIF} \\ \text{or} \\ C_{fr} \end{array}$$

where:

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C_{fms} = C_f from ASCE 7-95 Table 6-7 for moderately smooth type of surface (dimensionless)

C_{fr} = C_f from ASCE 7-95 Table 6-7 for rough type of surface (dimensionless)

WIF = Wind Increase Factor = $\frac{D + L_p + N_p}{D}$ (dimensionless)

L_p = Ladder projection (feet)

Note!!! In the absence of firm information, use $L_p = 1$ foot

N_p = Outside diameter of vapor nozzle, including insulation (feet)

D = Basic vessel diameter, equal to vessel inside diameter plus 2 times plate (wall) thickness plus 2 times insulation thickness (feet)

Note!!! In the absence of firm information, the following values of WIF may be used:

D (inches)	WIF
24 to 30	1.5
36 to 48	1.4
54 to 72	1.3
78 and greater	1.2

Platforms

Winds loads on platforms should be calculated for each platform and applied as a horizontal force at the platform elevation

$$F = (0.5) A q_z G$$

where:

F = Horizontal design wind force on platform (pounds)

A = Platform horizontal surface area (ft²)

q_z = Velocity pressure; determined at platform height z (psf)

G = Gust effect factor for vessel (dimensionless)

The arc of platform used to determine the platform area, A , should not exceed 180 degrees for any platform except for the platform at the top of the vessel.

The following criteria can be used to estimate the number and size of platforms. Review these criteria with the Piping Supervisor and adjust when required to meet contract requirements such as towers with many valves:

- One platform 2'- 6" below each manway for all manways 15 feet or greater above grade.
- One rectangular platform at the top of the vessel if required.

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- A minimum of one platform every 25 feet, extending around the vessel by the arc shown in the table below:

Vessel Diameter	Platform Arc, degrees
0" to 48"	180
49" to 96"	120
97" to 144"	90
145" and greater	60

Tall And Slender Vessels

The general procedure for vertical vessels requires the use of a modified gust effect factor G_f if the fundamental (first mode) frequency of vibration of the vessel is less than 1.0 Hertz or if the height to diameter ratio exceeds 4.0.

Fundamental Frequency

For a vessel with constant wall thickness, constant diameter, and a fixed base, the natural frequencies are those for a cantilever beam:

$$n_i = \frac{K_i}{H^2} \sqrt{\frac{EI}{m}}$$

where:

n_i	=	Frequency of mode i	(Hertz)
K_i	=	Constant	(dimensionless)
	=	0.560 for Mode 1	
	=	3.51 for Mode 2	
	=	9.82 for Mode 3	
	=	19.2 for Mode 4	

Note!!! Mode 1 is the only one required for calculating gust response factor. Modes 2, 3, and 4 may participate in across-wind response.

H	=	Height of vessel	(feet)
E	=	Modulus of elasticity	(psf)
I	=	Moment of inertia of vessel = $\frac{\pi d^3 t}{8}$	(ft ⁴)
m	=	Mass of vessel per unit length	(pounds-seconds ² /ft ²)
d	=	Inside vessel diameter	(feet)
t	=	Vessel wall thickness	(feet)

Note!!! For vertical vessels with variable diameter and/or wall thickness, more precise methods are available and may be appropriate. Consultation with the Vessels Engineer is recommended.

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Across-Wind Response

To evaluate the importance of across-wind response, calculate V_c and V_D as defined in Attachment 07. If $V_c > 1.3 V_D$, then across-wind response is not a concern. If $V_c < 1.3 V_D$, then evaluate the effects of across-wind response as described in Attachment 07 and ASME STS-1-1992.

Modified Gust Response Factor

Substitute G_f for G in the general procedure. Calculate G_f as detailed in ASCE 7-95 Commentary Section 6.6. This procedure requires the selection of an appropriate value for structural damping.

Structural Damping

The magnitude of the structural damping ratio, β , also called fraction of critical damping, depends not only on the vessel itself, but also on the vessel soil-structure interaction. Determination of damping values is not an exact science. Typical values are as follows:

Concrete vessel	0.0150 to 0.025
Steel vessel	0.0050 to 0.015
Unlined steel stack	0.0016 to 0.006
Gunitite-lined steel stack	0.0030 to 0.012
Concrete chimney	0.0040 to 0.020

The lower values are appropriate for foundation on rock or piles. Average values are appropriate for foundations on compacted soil. Higher values are appropriate for vessels supported by elevated structures or soft soils.

HORIZONTAL VESSELS

General Procedure

The following is derived from ASCE 7-95 Table 6-1 for "Main wind-force resisting systems" of "Open buildings and other structures":

$$F = q_z G C_r A_r$$

where:

F	=	Design wind force distribution on vessel	(pounds)
q_z	=	Velocity pressure, one value for entire vessel determined using z at vessel centerline height	(psf)
G	=	Gust effect factor	(dimensionless)
C_r	=	Force coefficient, one value for each wind direction. For wind parallel to the length of vessel, select value from ASCE 7-95 Table 6-7. For wind perpendicular to length	

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of vessel, select value from ASCE 7-95 Table 6-8.

Multiply this value by 0.7 to account for
cylindrical shape of vessel

(dimensionless)

Note!!! The use of Table 6-8 is appropriate because the wind flow perpendicular to the length of the horizontal vessel is divided above and below the vessel much as it would be by a billboard sign. The 0.7 factor accounts for the cylindrical shape of the vessel.

A_r = Projected area of vessel normal to the wind, one value for each wind direction. For wind along length of vessel, A_r equals 0.785 times D^2 . For wind perpendicular to length of vessel, A_r equals D times length of vessel (ft²)

D = Basic vessel diameter, equal to vessel inside diameter plus 2 times plate (wall) thickness plus 2 times insulation thickness (feet)

Wind On Appurtenances

The general procedure for horizontal vessels may require modification to account for vessel piers and for appurtenances such as ladders, piping, and platforms.

Piers

Calculate wind forces on vessel piers as described for individual columns.

Ladders And Piping

For wind along the length of the vessel, account for ladders and piping as described for vertical vessels.

For wind perpendicular to length of vessel, it is not necessary to account for those ladders and piping which are within the wind shadow of the vessel.

Platforms

Wind loads on platforms should be calculated for each platform and applied as a horizontal force at the platform elevation:

$$F = (0.5) A q_z G$$

where:

F = Horizontal design wind force on platform (pounds)

A = Platform horizontal surface area (ft²)

q_z = Velocity pressure, determined at platform height z (psf)

G = Gust effect factor for vessel (dimensionless)

Review platform requirements with the Piping Supervisor.

Sample calculations are given in Attachment 02.

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ENCLOSED STRUCTURES

For enclosed structures, ASCE 7-95 defines procedures for designing main wind-force resisting systems and components and cladding.

Note!!! Large roll-up doors near a corner of an enclosed structure may not have sufficient strength to resist local wind pressure. Consult with Door Manufacturer. If doors are not sufficiently strong, design the structure as "partially enclosed".

The general procedure for enclosed structures requires the use of a modified gust effect factor G_f if the fundamental (first mode) frequency of vibration of the structure is less than 1.0 Hertz or if the height to diameter ratio exceeds 4.0.

When calculating G_f , the value of structural damping should be selected as appropriate for the structural system; for example, 0.01 for bolted steel buildings and 0.02 for reinforced concrete buildings.

OPEN EQUIPMENT STRUCTURES

Open equipment structures support equipment and piping within an open structural frame, generally unenclosed by siding or other shielding appurtenances. Open equipment structures include:

- Open buildings as defined by ASCE 7-95 Section 6.2
- Pipe racks or cable tray racks
- Framed or trussed towers
- Structural frames supporting appurtenances

Procedures in this section are based on those recommended by ASCE *Wind Loads on Petrochemical Facilities*. Sample design calculations of an open building are given in Attachment 03 and of a pipe rack in Attachment 04.

General Procedure

$$F = q_z G C_f A_r$$

or

$$F = q_z G_f C_f A_r$$

where:

- | | | |
|-------|--|-----------------|
| F | = Design wind force distribution on main wind-force resisting system | (pounds) |
| q_z | = Velocity pressure, determined as varying with height | (psf) |
| G | = Gust effect factor | (dimensionless) |
| G_f | = Modified gust effect factor for flexible structures. Calculate as detailed in ASCE 7-95 Commentary Section 6.6 with appropriate values of structural | |

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damping, such as 0.01 for bolted steel structures and 0.02 for reinforced concrete ones. (dimensionless)

C_f = Force coefficient; as defined in this section (dimensionless)

A_f = Effective solid area; defined in the section Force Coefficients (ft²)

Note!!! It is not conservative to assume that an upper bound to wind force on an open structure is given by the force on that structure as if it were enclosed. ASCE *Wind Loads on Petrochemical Facilities* comments that model tests of open buildings have demonstrated that wind force on an open structure can exceed wind force on that structure when subsequently enclosed.

Force Coefficients

For open equipment structures which are square or nearly square in plan, use force coefficients from ASCE 7-95 Table 6-10 with solidity ratio ϵ as defined below.

For open equipment structures which are rectangular in plan and have flat-sided members, use force coefficients C_f as described below. (These coefficients are fit to ASCE 7-95 Table 6-10 and to ASCE *Wind Loads on Petrochemical Facilities* Figure 4.1.)

$$\text{For } N = 2 \text{ to } 4 \quad C_f = 1.8 + 1.4 N - (1.0 + 1.2 N) \epsilon^{0.45} \eta^{-0.06}$$

$$\text{For } N = 5 \text{ to } 7 \quad C_f = 3.0 + 1.2 N - (1.2 + 1.2 N) \epsilon^{0.45} \eta^{-0.02(N-1)}$$

where:

ϵ = Solidity ratio = A_f / A_g . Expressions above are based on data for $0.10 \leq \epsilon \leq 0.50$. For smaller solidity ratios, neglect shielding and use $C_f = 2.0$ for each member in each frame. For larger solidity ratios, use these expressions with caution. (dimensionless)

A_f = Effective solid area of frame, including beams, columns, bracing, cladding, stairs, ladders, handrail, horizontal projection of decking, etc. Do not include minor structural items, such as floor beams, which are not in the plane of a frame. Also, do not include items such as vessels, piping, or cable trays -- wind forces on these items are calculated separately as described elsewhere in this practice.

If all frames have equal solid area or if the windward frame has greater solid area than the others, use A_f as for the windward frame. If the solid area of the windward frame is less than that of some other frames, use A_f as the average of all frames. (ft²)

A_g = Gross area of the structure's envelope (ft²)

η = Frame spacing ratio = S_f / B . Expressions above are based on data for $0.10 \leq \eta \leq 0.50$ from ASCE *Wind Load on Petrochemical Facilities* and for $\eta = 1.0$ with $N = 2$

The value calculated is for all frames.

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from ASCE 7-95. They also agree well with test data reported by Whitbread for parallel trusses normal to wind. His data are for $2 \leq N \leq 5$ and $0.5 \leq \eta \leq 4.0$. (dimensionless)

S_f = Frame spacing center-to-center of frames, measured parallel to wind direction (feet)

B = Frame envelope width, measured normal to wind direction (feet)

N = Number of framing lines at spacing S_f . For $N > 7$, use curves in ASCE *Wind Loads on Petrochemical Facilities* Figure 4.1. (dimensionless)

Partially Sided Structures

For analysis of an open structure having siding on part of its surface, wind forces from the siding should be applied to the analysis model at siding support locations.

For modeling forces on the main wind-force resisting system, a force coefficient of 1.3, acting on the siding area, is appropriate.

If the siding extends around a corner or otherwise is subject to high local wind pressures, then design of the siding itself and its connections should be as for components and cladding in accordance with ASCE 7-95.

Shielding Of Equipment

It is conservative to calculate wind force on equipment in an open structure as if it is unshielded by either the structure or by other equipment, as described elsewhere in this practice. (See the section Other Considerations, Shielding.) If the engineer judges that there is significant shielding of equipment within an open structure, wind force as calculated elsewhere may be multiplied by a reduction factor, given by ASCE *Wind Loads on Petrochemical Facilities* as:

$$(1 - \epsilon)^{\kappa + 0.3} \quad \text{but not less than } 0.4$$

where:

ϵ = Solidity ratio defined previously (dimensionless)

κ = Volumetric solidity ratio for the floor level under consideration, defined as the ratio of the sum of the volumes of all the equipment on that level to the gross volume of the structure at that level.

κ should be taken as zero if there is only one item of equipment on the level, or if the equipment is widely spaced. (dimensionless)

Note!!! Do not reduce wind force on any portion of equipment which extends above the top of the structure.

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Modeling Wind Forces

It is important to apply wind forces to a structural analysis model so as to obtain realistic overturning and torsional effects. On the other hand, wind force calculations provide only an approximation to the forces a structure will see in a storm, and it is a waste of effort to be over-precise. Following are some guidelines, to be tempered with engineering judgment:

- For structures with frames having solidity ratio < 0.10 , apply wind forces to all frames. Otherwise, unless the windward frame has much less solidity than the others, apply wind forces to the windward frame.
- Wind reactions from equipment, partial siding, and concentrated piping should be located accurately to model overturning and torsional effects.
- Generally, it is sufficient to use one value of C_f per frame. An exception would be if the frame exhibits a significant variation in solidity.

Pipe Racks And Cable Tray Racks

Pipe racks or cable tray racks are specialized open equipment structures whose principal function is to support horizontal runs of piping, cable trays, or both.

Calculate wind forces on the structure as described above -- wind forces on piping and trays are calculated separately as described elsewhere in this practice.

If the rack is significantly longer than its width, only wind force in the transverse direction of the rack need be considered. For short racks with small pipe anchor loads, effects of longitudinal wind force should be evaluated.

Pipes

Wind loads on pipes are determined from the following, as recommended by ASCE *Wind Loads on Petrochemical Facilities*:

$$F = q_z G C_r (D + 0.1 W) L$$

where:

F	=	Design wind force on piping	(pounds)
q_z	=	Velocity pressure determined at z equal to piping height	(psf)
G	=	Gust effect factor as for the supporting structure	(dimensionless)
C_r	=	Force coefficient. Select from ASCE 7-95 Table 6-7 for round pipe having $h/D = 25$	(dimensionless)
D	=	Diameter of largest pipe	(feet)
W	=	Width of pipe rack	(feet)
L	=	Length of pipe tributary to pipe rack bent	(feet)

Note!!! The procedure described above for wind load on pipes assumes that wind approaches at an angle of up to 6 degrees from the horizontal and that the largest pipe shields the others. Engineering judgment must be used to determine whether

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this model is appropriate. If, for example, there are large pipes separated by several diameters, it may be appropriate to apply wind load to each of them.

Note!!! Trussed towers and multi-level open buildings are likely to have vertical runs of piping. If piping arrangements within such a structure are unknown, assume that pipe covers 10% of the structure's gross area for each wind approach direction, and use $C_f = 0.7$.

Cable Trays

Wind loads on cable trays are determined from the following, as recommended by ASCE *Wind Loads on Petrochemical Facilities*:

$$F = q_z G C_f (D + 0.1 W) L$$

where:

F	=	Design wind force on trays	(pounds)
q_z	=	Velocity pressure determined at z equal to tray height	(psf)
G	=	Gust effect factor as for the supporting structure	(dimensionless)
C_f	=	Force coefficient = 2.0	(dimensionless)
D	=	Depth of deepest tray	(feet)
W	=	Width of rack	(feet)
L	=	Length of tray tributary to one bent	(feet)

Note!!! The procedure described above for wind load on trays assumes that wind approaches at an angle of up to 6 degrees from the horizontal and that the windward tray shields the others. Engineering judgment must be used to determine whether this model is appropriate. If, for example, there are large separations between trays, it may be appropriate to apply wind load to each of them.

Appurtenances

Wind loads on structural members supporting appurtenances should be determined as for open equipment structures.

Wind loads on air coolers should be determined as for enclosed structures, except do not consider uplift forces on air coolers.

INDIVIDUAL COLUMNS

Individual columns are cantilever columns supporting utilities, platforms, or vessels. Tee supports should be considered as individual columns. A sample design is given in Attachment 05.

Wind loads on individual columns are determined from the following formula:

$$F = q_h G C_f A_f$$

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where:

F	=	Design wind force on column	(pounds)
q_h	=	Velocity pressure determined at z equal to column height	(psf)
G	=	Gust effect factor	(dimensionless)
C_f	=	Force coefficient as follows:	(dimensionless)
		For flat-sided shapes, $C_f = 2.0$	
		For round shapes, use C_f from ASCE 7-95 Table 6-7	
A_f	=	Tributary area normal to wind direction	(ft ²)

Wind On Appurtenances

Wind on ladders, piping, cable trays, and platforms supported by individual columns should be determined as for vertical vessels.

LOAD COMBINATIONS

Use load combinations from ASCE 7-95 Section 2 and Structural Engineering Specification 670.215.00910, *Structural Engineering Criteria*, unless applicable local codes or Client requires otherwise.

Enclosed Structures

For wind loads on enclosed structures, use full and partial loadings as described in ASCE 7-95 Section 6.8.

Open Equipment Structures

For wind loads on open equipment structures, calculate the sum of wind loads on the structure, on equipment, and on piping and cable trays for wind directions parallel to each primary axis of the structure.

For open equipment structures which are square or nearly square in plan, analyze at least two wind directions:

- Normal to a face
- On a diagonal. Follow notes in ASCE 7-95 Table 6-10 for calculating diagonal wind forces on the structure

For open equipment structures rectangular in plan, analyze at least two wind force load combinations:

- Full longitudinal wind force with 50% of transverse wind force
- Full transverse wind force with 50% of longitudinal wind force

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These cases are recommended in ASCE *Wind Loads on Petrochemical Facilities*. It notes that the 50%-value is an approximation to the force acting on the secondary axis, and it provides a more detailed method of calculating that force.

This secondary force must be considered because, for an open structure with more than one frame, the maximum wind force normal to a face occurs when the wind direction is somewhat oblique to that face. (For oblique winds, there is less shielding of successive columns by one another, and there is a wider width of the structure exposed directly to the wind.) Consequently, the wind direction which causes maximum load on one set of frames also causes significant load in frames perpendicular to those.

The secondary force may be neglected in the following circumstances:

- When the full force along one axis is considerably greater than along the other, as for a long pipe rack
- When the solidity ratio ϵ is less than 0.10, and shielding is neglected with $C_r = 2.0$ used for wind force calculations on each member

OTHER CONSIDERATIONS

Drift Control

As with earthquake design, lateral drift limits must be considered in wind design. Unlike with earthquake design, there are no code-prescribed drift limits corresponding to the prescribed design wind forces. ASCE published a state-of-the art report in 1988 addressing wind drift design. This report recommends that wind drift for enclosed buildings be limited to Height / 400 for a 10-year return period wind. ASCE 7-95 Commentary Table C6-5 provides conversion factors among wind speeds having various return periods.

PIP STC 01015 addresses allowable drift limits for structures in petrochemical facilities, and provides for the following limits:

- | | |
|--|---|
| ■ For pipe racks | Height / 150 |
| ■ For process structures, pre-engineered metal buildings, and personnel access platforms | Height / 200 |
| ■ For structures with bridge cranes | The smaller of 2 inches or Height / 200 |
| ■ For occupied buildings which may be damaged by excessive drift | Height / 400 |

Overturning Stability

The overturning moment due to wind load should not exceed 2/3 of the resisting moment of the structure during its lightest possible weight condition after plant construction has been completed.

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Shielding

No reduction in wind loads shall be made for the shielding effects of vessels or structures adjacent to the one being designed. ASCE 7-95 Section 6.5.4 does not permit consideration of possible shielding of one building or structure by another unless verified by tests.

REFERENCES

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PIP (Process Industry Practices) STC 01015. *Structural Design Criteria*. Draft, 1998.

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ATTACHMENTS

Attachment 01:

Sample Design 1: Wind Load -- Vertical Vessel

Attachment 02:

Sample Design 2: Wind Load -- Horizontal Vessel

Attachment 03:

Sample Design 3: Wind Load -- Open Equipment Structure

Attachment 04:

Sample Design 4: Wind Load -- Pipe Rack

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Attachment 05:
Sample Design 5: Wind Load -- Tee Support Column

Attachment 06:
General Discussion

Attachment 07:
Across-Wind Response

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SAMPLE DESIGN 1 - VERTICAL VESSEL

GIVEN:

Ponca City, Oklahoma

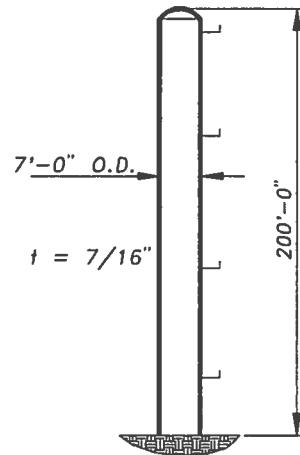
Oil Refinery

Vertical Vessel

No Insulation

Platforms, 60°, 3 ft. wide at 15, 75, 125, and 190 ft.

Damping, $\beta = 0.01$



REQUIRED:

Wind forces on empty vessel

SOLUTION:

Determine Velocity Pressure

Oil Refinery is in Building Category III

{ASCE 7-95, Table 1-1}

Importance Factor, $I = 1.15$ for Building Category III

{ASCE 7-95, Table 6-2}

Exposure Category C for open terrain

{ASCE 7-95, Section 6.5.3}

Basic Wind Speed, $V = 90$ mph

{ASCE 7-95, Figure 6-1}

$$q_z = 0.00256 K_z K_{zt} V^2 I = 0.00256 K_z (1.00) (90)^2 (1.15) = 23.8 K_z \text{ psf}$$

K_z for Exposure Category C

{ASCE 7-95, Table 6-3}

Height	K_z	q_z
200 ft	1.46	34.7 psf
180 ft	1.43	34.0 psf
160 ft	1.39	33.1 psf
140 ft	1.36	32.4 psf
120 ft	1.31	31.2 psf
100 ft	1.26	30.0 psf
90 ft	1.24	29.5 psf
80 ft	1.21	28.8 psf
70 ft	1.17	27.8 psf
60 ft	1.13	26.9 psf
50 ft	1.09	25.9 psf
40 ft	1.04	24.8 psf
30 ft	0.98	23.3 psf
25 ft	0.94	22.4 psf
20 ft	0.90	21.4 psf
15 ft	0.85	20.2 psf

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SAMPLE DESIGN 1 - VERTICAL VESSEL

Determine Fundamental Frequency

$$I = \frac{\pi d^3 t}{8} = \frac{\pi (6.93 \text{ ft})^3 (0.0365 \text{ ft})}{8} = 4.76 \text{ ft}^4$$

$$m = \frac{\pi D t \gamma}{g} = \frac{\pi (7 \text{ ft}) (0.0365 \text{ ft}) (0.490 \text{ K/ft}^3)}{32.2 \text{ ft/sec}^2} = 0.0122 \text{ K-sec}^2/\text{ft}^2$$

$$n_1 = \frac{K_1 \sqrt{\frac{EI}{m}}}{H^2} = \frac{0.560}{(200 \text{ ft})^2} \sqrt{\frac{(29000 \text{ K/in}^2) (4.76 \text{ ft}^4) (144 \text{ in}^2/\text{ft}^2)}{(0.0122 \text{ K-sec}^2/\text{ft}^2)}} = 0.565 \text{ Hz} < 1.0 \text{ H}$$

Determine Gust Response Factor, G_r

Values From ASCE 7-95, Table C6-6 For Exposure C

$$\bar{\alpha} = \frac{1}{6.5} = 0.1538$$

$$\bar{b} = 0.65$$

$$c = 0.20$$

$$l = 500 \text{ ft}$$

$$\varepsilon = \frac{1}{5.0} = 0.20$$

$$z_{\min} = 15 \text{ ft}$$

Values From Vessel Geometry

$$h = 200 \text{ ft}$$

$$b = 7 \text{ ft}$$

$$D = 7 \text{ ft}$$

Calculated Values

$$\bar{z} = 0.6h = 0.6(200 \text{ ft}) = 120 \text{ ft} \quad (> z_{\min} = 15 \text{ ft } \underline{\text{ok}}) \quad \{\text{ASCE Table C6-6}\}$$

$$I_z = c \left(\frac{33}{\bar{z}} \right)^{\frac{1}{6}} = 0.20 \left(\frac{33 \text{ ft}}{120 \text{ ft}} \right)^{.167} = 0.161 \quad \{\text{ASCE 7-95, Eq. C6-6}\}$$

$$L_z = l \left(\frac{\bar{z}}{33} \right)^{\varepsilon} = 500 \text{ ft} \left(\frac{120 \text{ ft}}{33 \text{ ft}} \right)^{0.20} = 647 \text{ ft} \quad \{\text{ASCE 7-95, Eq. C6-8}\}$$

$$Q^2 = \frac{1}{1 + 0.63 \left(\frac{b+h}{L_z} \right)^{0.63}} = \frac{1}{1 + 0.63 \left(\frac{7+200 \text{ ft}}{647 \text{ ft}} \right)^{0.63}} = 0.764 \quad \{\text{ASCE 7-95, Eq. C6-7}\}$$

$$\hat{V}_{ref} = \frac{(90 \text{ miles/hour})(5280 \text{ ft/mile})}{(3600 \text{ sec/hour})} = 132 \text{ ft/se}$$

FLUOR DANIEL

SAMPLE DESIGN 1 - VERTICAL VESSEL

$$\bar{V}_z = \bar{b} \left(\frac{\bar{z}}{33} \right)^{\bar{\alpha}} \left(\hat{V}_{ref} \right) = 0.65 \left(\frac{120 \text{ ft}}{33 \text{ ft}} \right)^{0.1538} (132 \text{ ft/sec}) = 104.6 \text{ ft/sec}$$

$$N_1 = \frac{n_1 L_z}{\bar{V}_z} = \frac{0.57 \text{ Hz}(647 \text{ ft})}{104.6 \text{ ft/sec}} = 3.53$$

$$\eta_h = \frac{4.6 n_1 h}{\bar{V}_z} = \frac{4.6(0.57 \text{ Hz})(200 \text{ ft})}{104.6 \text{ ft/sec}} = 5.01$$

$$\eta_b = \frac{4.6 n_1 b}{\bar{V}_z} = \frac{4.6(0.57 \text{ Hz})(7 \text{ ft})}{104.6 \text{ ft/sec}} = 0.175$$

$$\eta_d = \frac{15.4 n_1 D}{\bar{V}_z} = \frac{15.4(0.57 \text{ Hz})(7 \text{ ft})}{104.6 \text{ ft/sec}} = 0.587$$

$$R_n = \frac{7.465 N_1}{(1 + 10.302 N_1)^{\frac{2}{3}}} = \frac{7.465(3.53)}{[1 + 10.302(3.53)]^{1.667}} = 0.063$$

$$R_h = \frac{1}{\eta_h} - \frac{1}{2\eta_h^2} (1 - e^{-2\eta_h}) = \frac{1}{5.01} - \frac{1}{2(5.01)^2} (1 - e^{-2(5.01)}) = 0.199 - 0.0199(1 - 0.0000445) = 0.179$$

$$R_b = \frac{1}{\eta_b} - \frac{1}{2\eta_b^2} (1 - e^{-2\eta_b}) = \frac{1}{0.175} - \frac{1}{2(0.175)^2} (1 - e^{-2(0.175)}) = 5.714 - 16.326(1 - 0.705) = 0.898$$

$$R_d = \frac{1}{\eta_d} - \frac{1}{2\eta_d^2} (1 - e^{-2\eta_d}) = \frac{1}{0.587} - \frac{1}{2(0.587)^2} (1 - e^{-2(0.587)}) = 1.704 - 1.451(1 - 0.309) = 0.701$$

$$R^2 = \frac{1}{\beta} R_n R_h R_b (0.53 + 0.47 R_d) = \frac{1}{0.01} (0.063)(0.179)(0.898)[0.53 + 0.47(0.701)] = 0.87$$

$$G_f = \frac{1 + 2gI_z \sqrt{Q^2 + R^2}}{1 + 7I_z} = \frac{1 + 2(3.5)(0.161) \sqrt{0.764 + 0.870}}{1 + 7(0.161)} = \frac{2.441}{2.127} = 1.1 \quad \{\text{ASCE 7-95, Eq. C6-9}\}$$

Determine Pressure Coefficient, C_r

{ASCE 7-97, Table 6-7}

$$\text{WIF} = 1.2 \text{ for } D = 7'-0" = 84"$$

$$D \sqrt{q_z} = 7 \sqrt{34.7} = 41.2 > 2.$$

$$\frac{h}{D} = \frac{200 \text{ ft}}{7 \text{ ft}} = 28.6$$

For moderately smooth surface: $C_r = 0.7$

For rough surface: $C_r = 0.9$

$$C_r = 0.7(1.2) = 0.84$$

governs

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SAMPLE DESIGN 1 - VERTICAL VESSEL

Determine Pressure Forces On Platforms

$$F = (0.5)Aq_zG$$

$$A = \frac{\pi}{4}[(13 \text{ ft})^2 - (7 \text{ ft})^2] \frac{60^\circ}{360^\circ} = 15.7 \text{ ft}^2$$

$$G = 0.85 \text{ for Exposure Category C}$$

{ASCE 7-95, Section 6.6.1}

Elevation	Constant	A	q_z	G	F
190 ft	0.50	15.7 ft ²	34.3 psf	0.85	229# @ 190 ft
125 ft	0.50	15.7 ft ²	31.5 psf	0.85	210# @ 125 ft
75 ft	0.50	15.7 ft ²	28.3 psf	0.85	189# @ 75 ft
15 ft	0.50	15.7 ft ²	20.2 psf	0.85	135# @ 15 ft

Determine Pressure Forces On Vessel

$$F = q_z G_f C_f A_f$$

$$A_f = 7 \text{ ft} \times \text{tributary height}$$

Elevation	q_z	G_f	C_f	Trib. Ht.	A_f	F
190-200 ft	34.7 psf	1.15	0.90	10 ft	70 ft ²	2514# @ 195 ft
170-190 ft	34.0 psf	1.15	0.90	20 ft	140 ft ²	4927# @ 180 ft
150-170 ft	33.1 psf	1.15	0.90	20 ft	140 ft ²	4796# @ 160 ft
130-150 ft	32.4 psf	1.15	0.90	20 ft	140 ft ²	4695# @ 140 ft
110-130 ft	31.2 psf	1.15	0.90	20 ft	140 ft ²	4521# @ 120 ft
95-110 ft	30.0 psf	1.15	0.90	15 ft	105 ft ²	3260# @ 103 ft
85-95 ft	29.5 psf	1.15	0.90	10 ft	70 ft ²	2137# @ 90 ft
75-85 ft	28.8 psf	1.15	0.90	10 ft	70 ft ²	2087# @ 80 ft
65-75 ft	27.8 psf	1.15	0.90	10 ft	70 ft ²	2014# @ 70 ft
55-65 ft	26.9 psf	1.15	0.90	10 ft	70 ft ²	1949# @ 60 ft
45-55 ft	25.9 psf	1.15	0.90	10 ft	70 ft ²	1876# @ 50 ft
35-45 ft	24.8 psf	1.15	0.90	10 ft	70 ft ²	1797# @ 40 ft
27.5-35 ft	23.3 psf	1.15	0.90	7.5 ft	53 ft ²	1278# @ 32 ft
22.5-27.5 ft	22.4 psf	1.15	0.90	5 ft	35 ft ²	811# @ 25 ft
17.5-22.5 ft	21.4 psf	1.15	0.90	5 ft	35 ft ²	775# @ 20 ft
0-17.5 ft	20.2 psf	1.15	0.90	17.5 ft	123 ft ²	2571# @ 9 ft

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SAMPLE DESIGN 1 - VERTICAL VESSEL

Check Across-Wind Response

$$V_c = \frac{(0.57 \text{ Hz})(7 \text{ ft})}{0.2} = 20.0 \text{ ft/sec}$$

$$V_D = 0.96(90 \text{ mph})\sqrt{1.46} = 104 \text{ ft/se}$$

$$0.2V_D = 0.2(104 \text{ ft/sec}) = 20.8 \text{ ft/sec}$$

$$0.4V_D = 0.4(104 \text{ ft/sec}) = 41.6 \text{ ft/sec}$$

$$1.3V_D = 1.3(104 \text{ ft/sec}) = 135 \text{ ft/sec}$$

$$M = \frac{m\beta}{\rho D^2} = \frac{(0.0122 \text{ K-sec}^2/\text{ft}^2)(0.01)}{(0.0000024 \text{ K-sec}^2/\text{ft}^4)(7 \text{ ft})^2} = 1.04$$

$$M > 0.8 \text{ and } V_c < 0.2V_D$$

1st mode OK, Check 2nd mode

Check 2nd Mode

$$n_2 = (0.57 \text{ Hz})\left(\frac{3.534}{0.560}\right) = 3.60 \text{ Hz}$$

$$V_c = \frac{(3.60 \text{ Hz})(7 \text{ ft})}{0.2} = 126 \text{ ft/sec}$$

$$M > 0.8 \text{ and } 0.4V_D < V_c < 1.3V_D$$

Refer to the ASME standard for further guidance

FLUOR DANIEL

SAMPLE DESIGN 2 - HORIZONTAL VESSEL

GIVEN:

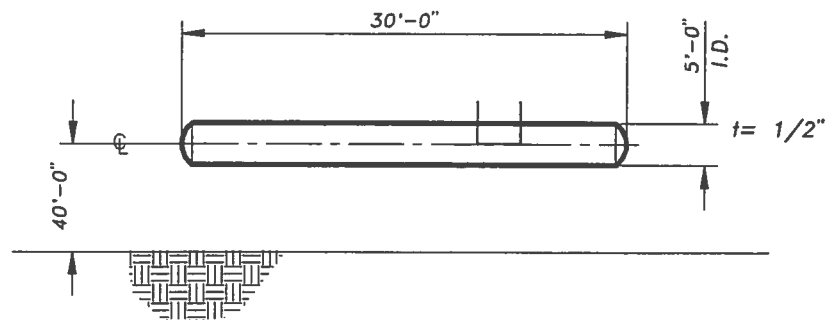
Galveston, Texas

Oil Refinery

Horizontal Vessel

2" Insulation

2 platforms at centerline, 3 feet wide,
6 feet long



REQUIRED:

Wind forces on vessel

SOLUTION:

Determine Velocity Pressure

Oil Refinery is in Building Category III

{ASCE 7-95, Table 1-1}

Importance Factor, $I = 1.15$ for Building Category III

{ASCE 7-95, Table 6-2}

Exposure Category D for Texas coastline

{ASCE 7-95, Section 6.5.3}

Basic Wind Speed, $V = 125$ mph

{ASCE 7-95, Figure 6-1}

$K_z = 1.22$ for Exposure Category D at 40 feet

{ASCE 7-95, Table 6-3}

$$q_z = 0.00256 K_z K_d V^2 I = 0.00256(1.22)(1.00)(125)^2(1.15) = 56.1 \text{ psf}$$

Determine Pressure Coefficients, C_f

Longitudinal Wind

{ASCE 7-95, Table 6-7}

$$D = 60" + 2(1/2" + 2") = 65"$$

WIF = 1.3 for $D = 65"$

$$D \sqrt{q_z} = 5.42 \sqrt{56.1} = 40.6 > 2.$$

$$\frac{h}{D} = \frac{5.42 \text{ ft}}{5.42 \text{ ft}} = 1.0$$

For moderately smooth surface: $C_f = 0.5$

For rough surface: $C_f = 0.7$

governs

$$C_f = 1.3(0.5) = 0.65$$

Transverse Wind

{ASCE 7-95, Table 6-8}

$$\frac{M}{N} = \frac{30 \text{ ft}}{5.42 \text{ ft}} = 5.53$$

$$C_f = 0.7(1.2) = 0.84$$

FLUOR DANIEL

SAMPLE DESIGN 2 - HORIZONTAL VESSEL

Determine Pressure Forces On Platforms

$$F = (0.5)Aq_zG$$

$$G = 0.85 \text{ for Exposure Category D}$$

{ASCE 7-95, Section 6.6.1}

$$A = (3 \text{ ft})(6 \text{ ft}) = 18 \text{ ft}^2$$

$$F = (0.5)(18 \text{ ft}^2)(56.1 \text{ psf})(0.85) = 429\#, \text{ each platform, each direction}$$

Determine Pressure Forces On Vessel

$$F = q_zGC_rA_r$$

Longitudinal Wind

$$A_f = 0.785(5.42 \text{ ft})^2 = 23.1 \text{ ft}^2$$

$$F = (56.1 \text{ psf})(0.85)(0.7)(23.1 \text{ ft}^2) = 771\#$$

Transverse Wind

$$A_r = (30 \text{ ft})(5.42 \text{ ft}) = 163 \text{ ft}^2$$

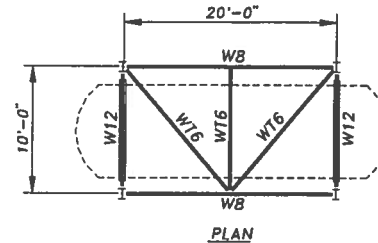
$$F = (56.1 \text{ psf})(0.85)(0.84)(163 \text{ ft}^2) = 6,529\#$$

FLUOR DANIEL

SAMPLE DESIGN 3 - OPEN EQUIPMENT STRUCTURE

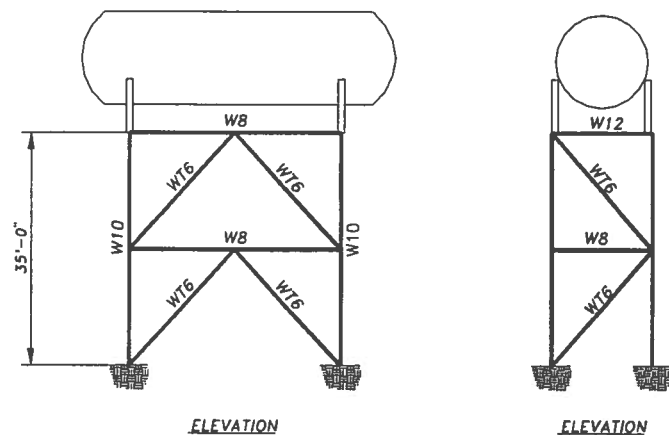
GIVEN:

Galveston, Texas
Oil Refinery
Supports horizontal vessel (See Sample Design 2)
No platforms
Minimal piping



REQUIRED:

Wind forces on structure



SOLUTION:

Determine Velocity Pressure

Oil Refinery is in Building Category III

{ASCE 7-95, Table 1-1}

Importance Factor, $I = 1.15$ for Building Category III

{ASCE 7-95, Table 6-2}

Exposure Category D for Texas coastline

{ASCE 7-95, Section 6.5.3}

Basic Wind Speed, $V = 125$ mph

{ASCE 7-95, Figure 6-1}

$$q_z = 0.00256 K_z K_{zt} V^2 I = 0.00256 K_z (1.00) (125)^2 (1.15) = 46.0 K_z \text{ psf}$$

K_z for Exposure Category D

{ASCE 7-95, Table 6-3}

Height	K_z	q_z
35 ft	1.19	54.7 psf
17.5 ft	1.05	48.3 psf
0 ft	1.03	47.4 psf

Determine Pressure Coefficients, C_p

Longitudinal Frame

$$A_{\text{tributary to 35 ft}} = 1(1 \text{ ft})(10 \text{ ft}) + 2(0.83 \text{ ft})(8.8 \text{ ft}) + 1(0.50 \text{ ft})(10.1 \text{ ft}) = 10.0 + 14.6 + 5.0 = 29.6 \text{ ft}^2$$

$$A_{\text{tributary to 17.5 ft}} = 1(0.67 \text{ ft})(10 \text{ ft}) + 2(0.83 \text{ ft})(17.5 \text{ ft}) + 1(0.50 \text{ ft})(20.2 \text{ ft}) = 6.7 + 29.0 + 10.1 = 45.8 \text{ ft}^2$$

$$A_{\text{tributary to grade}} = 2(0.83 \text{ ft})(8.8 \text{ ft}) + 1(0.50 \text{ ft})(10.1 \text{ ft}) = 14.6 + 5.0 = 19.6 \text{ ft}^2$$

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SAMPLE DESIGN 3 - OPEN EQUIPMENT STRUCTURE

$$A_{\text{projected}} = (10 \text{ ft} + 0.83 \text{ ft})(35 \text{ ft} + 0.33 \text{ ft}) = 383 \text{ ft}^2$$

$$\varepsilon = \frac{(29.6 + 45.8 + 19.6)}{383} = 0.25$$

$$0.10 \leq \varepsilon \leq 0.50$$

$$N = 2$$

$$\eta = \frac{S_f}{B} = \frac{20}{10} = 2.0$$

$$C_f = 1.8 + 1.4(2) - [1.0 + 1.4(2)]0.25^{0.45}2.0^{-0.06} = 2.85$$

Transverse Frame

$$A_{\text{tributary to 35 ft}} = 1(0.67 \text{ ft})(20 \text{ ft}) + 2(0.83 \text{ ft})(8.8 \text{ ft}) + 2(0.50 \text{ ft})(10.1 \text{ ft}) = 13.4 + 14.6 + 10.1 = 38.1 \text{ ft}^2$$

$$A_{\text{tributary to 17.5 ft}} = 1(0.67 \text{ ft})(20 \text{ ft}) + 2(0.83 \text{ ft})(17.5 \text{ ft}) + 2(0.50 \text{ ft})(20.2 \text{ ft}) = 13.4 + 29.0 + 20.2 = 62.6 \text{ ft}^2$$

$$A_{\text{tributary to grade}} = 2(0.83 \text{ ft})(8.8 \text{ ft}) + 2(0.50 \text{ ft})(10.1 \text{ ft}) = 14.6 + 10.1 = 24.7 \text{ ft}^2$$

$$A_{\text{projected}} = (20 \text{ ft} + 0.83 \text{ ft})(35 \text{ ft} + 0.33 \text{ ft}) = 736 \text{ ft}^2$$

$$\varepsilon = \frac{(38.1 + 62.6 + 24.7)}{736} = 0.17$$

$$0.10 \leq \varepsilon \leq 0.50$$

$$N = 2$$

$$\eta = \frac{10}{20} = 0.5$$

$$C_f = 1.8 + 1.4(2) - [1.0 + 1.2(2)]0.17^{0.45}0.5^{-0.06} = 3.00$$

Determine Pressure Forces On Structure

$$F = q_z G C_f A_r$$

$$G = 0.85 \text{ for Exposure Category D}$$

{ASCE 7-95, Section 6.6.1}

Wind On Longitudinal Frame

$$\text{At 35 ft: } F = (54.7 \text{ psf})(0.85)(2.85)(29.6 \text{ ft}^2) = 3,922\#$$

$$\text{At 17.5 ft: } F = (48.3 \text{ psf})(0.85)(2.85)(45.8 \text{ ft}^2) = 5,359\#$$

$$\text{At 0 ft: } F = (47.4 \text{ psf})(0.85)(2.85)(19.6 \text{ ft}^2) = 2,251\#$$

Wind On Transverse Frame

$$\text{At 35 ft: } F = (54.7 \text{ psf})(0.85)(3.00)(38.1 \text{ ft}^2) = 5,314\#$$

$$\text{At 17.5 ft: } F = (48.3 \text{ psf})(0.85)(3.00)(62.6 \text{ ft}^2) = 7,710\#$$

$$\text{At 0 ft: } F = (47.4 \text{ psf})(0.85)(3.00)(24.7 \text{ ft}^2) = 2,985\#$$

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SAMPLE DESIGN 4 - PIPE RACK

GIVEN:

Marcus Hook, PA. (South Of Philadelphia)

Oil Refinery

Unstrutted Pipeway

Steel Frames, 20 foot spacing

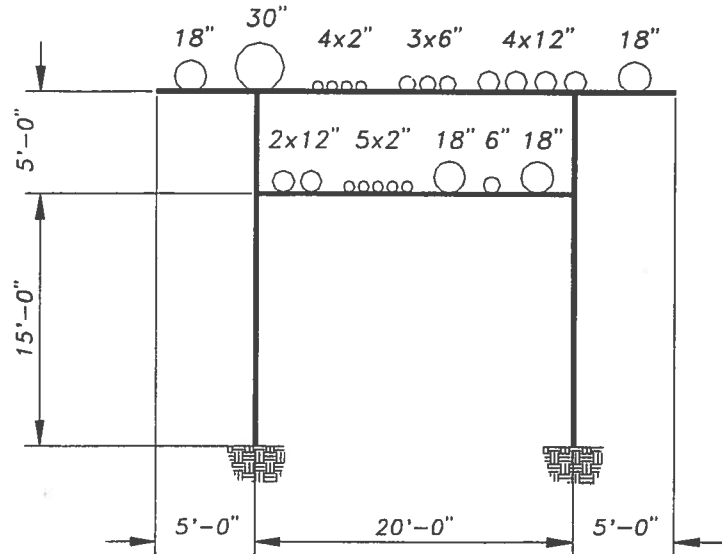
W14 Columns

No Platforms

No Cable Trays

REQUIRED:

Transverse Wind Forces On Pipe Rack



ELEVATION

SOLUTION:

Determine Velocity Pressure

Oil Refinery is in Building Category III

{ASCE 7-95, Table 1-1}

Importance Factor, $I = 1.15$ for Building Category III

{ASCE 7-95, Table 6-2}

Exposure Category C for open terrain

{ASCE 7-95, Section 6.5.3}

Basic Wind Speed, $V = 105$ mph

{ASCE 7-95, Figure 6-1}

$$q_z = 0.00256 K_z K_{zt} V^2 I = 0.00256 K_z (1.00) (105)^2 (1.15) = 32.5 K_z \text{ psf}$$

K_z for Exposure Category C

{ASCE 7-95, Table 6-3}

Height	K_z	q_z
20 ft	0.90	29.2 psf
15 ft	0.90	27.6 psf

Determine Gust Effect Factor

$G = 0.85$ for Exposure Category C

{ASCE 7-95, Section 6.6.1}

Determine Pressure Forces On Piping

Top Pipe Level

$D = 30 \text{ in} = 2.5 \text{ ft}$

for pipe racks, use $\frac{h}{D} = 25$

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SAMPLE DESIGN 4 - PIPE RACK

$C_r = 0.7$ for moderately smooth surface

{ASCE 7-95, Table 6-7}

$$F = q_z GC_r (D + 0.1W)L = (29.2 \text{ psf})(0.85)(0.7)[2.5 \text{ ft} + 0.1(30 \text{ ft})](20 \text{ ft}) = 1911\#$$

Lower Pipe Level

$$D = 18 \text{ in} = 1.5 \text{ ft}$$

- for pipe racks, use $\frac{h}{d} = 25$

$C_r = 0.7$ for moderately smooth surface

{ASCE 7-95, Table 6-7}

$$F = q_z GC_r (D + 0.1W)L = (27.6 \text{ psf})(0.85)(0.7)[1.5 \text{ ft} + 0.1(20 \text{ ft})](20 \text{ ft}) = 1150\#$$

Determine Pressure Forces On Structure

$$\varepsilon = \frac{(1.17 \text{ ft})(20 \text{ ft})}{(20 \text{ ft})(20 \text{ ft})} = 0.06 \leq 0.1$$

therefore, neglect shielding

$C_r = 2.0$ for first and second planes

Tributary To Top Pipe Level

$$A_r = (1.17 \text{ ft})(2.5 \text{ ft}) = 2.93 \text{ ft}^2$$

$$F = q_z GC_r A_r = (29.2 \text{ psf})(0.85)(2.0 + 2.0)(2.93 \text{ ft}^2) = 291\#$$

Tributary To Lower Pipe Level

$$A_r = (1.17 \text{ ft})(10 \text{ ft}) = 11.7 \text{ ft}^2$$

$$F = q_z GC_r A_r = (27.6 \text{ psf})(0.85)(2.0 + 2.0)(11.7 \text{ ft}^2) = 1098\#$$

Tributary To Foundation Level

$$A_r = (1.17 \text{ ft})(7.5 \text{ ft}) = 8.78 \text{ ft}^2$$

$$F = q_z GC_r A_r = (27.6 \text{ psf})(0.85)(2.0 + 2.0)(8.78 \text{ ft}^2) = 824\#$$

FLUOR DANIEL

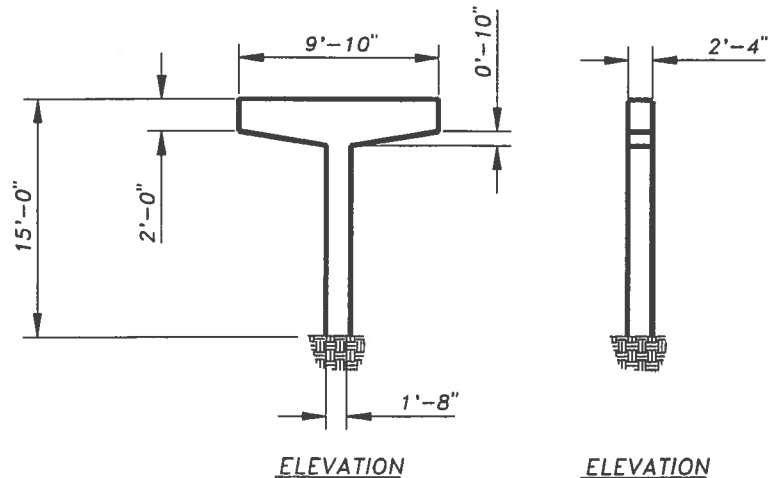
SAMPLE DESIGN 5 - TEE SUPPORT COLUMN

GIVEN:

Chillicothe, Ohio. (South Of Columbus)
Oil Refinery
Concrete

REQUIRED:

Wind Forces On Tee Support For
Horizontal Vessel



SOLUTION:

Determine Velocity Pressure

Oil Refinery is in Building Category III {ASCE 7-95, Table 1-1}
Importance Factor, $I = 1.15$ for Building Category III {ASCE 7-95, Table 6-2}
Exposure Category C for open terrain {ASCE 7-95, Section 6.5.3}
Basic Wind Speed, $V = 90$ mph {ASCE 7-95, Figure 6-1}
 $q_z = 0.00256 K_z K_{xt} V^2 I = 0.00256 K_z (1.00) (90)^2 (1.15) = 23.8 K_z \text{ psf}$
 K_z for Exposure Category C {ASCE 7-95, Table 6-3}

Height	K_z	q_z
15 ft	0.85	20.2 psf

Determine Pressure Forces On Structure

$G = 0.85$ for Exposure Category C {ASCE 7-95, Section 6.6.1}

$C_f = 2.0$ for flat sided shapes

$$F = q_h G C_f A_f$$

Vessel Longitudinal Direction (1'-8" Column Face)

$$\text{At 14 ft: } F = (20.2 \text{ psf})(0.85)(2.0)(9.83 \text{ ft} \times 2.0 \text{ ft}) = 675\#$$

$$\text{At 12.72 ft: } F = (20.2 \text{ psf})(0.85)(2.0)(.5 \times 9.83 \text{ ft} \times 0.83 \text{ ft}) = 140\#$$

$$\text{At 6.08 ft: } F = (20.2 \text{ psf})(0.85)(2.0)(1.67 \text{ ft} \times 12.17 \text{ ft}) = 698\#$$

Vessel Transverse Direction (2'-4" Column Face)

$$\text{At 7.5 ft: } F = (20.2 \text{ psf})(0.85)(2.0)(2.33 \text{ ft} \times 15.0 \text{ ft}) = 1,200\#$$

FLUOR DANIEL

GENERAL DISCUSSION

WIND CHARACTERISTICS

For structural design purposes, it is important to understand winds near the ground surface. The ensuing discussion on wind characteristics focuses on surface winds: the winds at 10 meters (33 feet) height above ground.

General Procedure

Most building codes define wind pressures and forces using equations of one or both of the following forms:

$$F = qGCA$$

or

$$P_w = qGC$$

where

F	=	Design wind force in pounds, acting in direction of wind.
P _w	=	Design wind pressure in pounds per square feet; positive value means acting towards the surface; negative value means acting away from the surface.
q	=	Velocity pressure in pounds per square feet.
G	=	Gust response factor (dimensionless).
C	=	Pressure coefficient (dimensionless).
A	=	Areas of structure projected normal to the wind in square feet.

Wind Speed And Velocity

Design wind speed depends on wind climate at a geographic location. Wind speed is usually determined on a probabilistic basis. Most design wind speeds in the United States are specified with an annual probability of exceedance of 0.02 (50 year mean recurrence interval). In addition to wind climate, wind speed depends on terrain over which the wind passes and on height above ground.

Variation Of Wind Speed With Height

Local wind speed is zero at the ground surface and it increases with height above ground within the atmosphere boundary layer. Above this layer exists the gradient wind, which does not vary with height. Wind speed within the boundary layer can be approximated by the equation:

$$V_z = V_g \left(\frac{Z}{z_g} \right)^{\frac{1}{\alpha}}$$

where:

V _z	=	Velocity (wind speed) at height Z.
V _g	=	Velocity (wind speed) at gradient height z _g .
α	=	Power law exponent that depends on surface roughness.

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GENERAL DISCUSSION

Variation Of Wind Speed With Surface Roughness

The rougher the terrain is, the more retarded the wind in the atmospheric boundary layer. In general, the rougher the terrain is, the higher the values of the gradient height, z_g , and the power law exponent $1/\alpha$ are, and the smaller the velocity, V_z , is at height Z .

The values used in ASCE 7-95, Table C6-2, are typical for United States Building Codes and are as follows:

Exposure Category	z_g	α
A Large Cities	1500 ft	5.0
B Urban and Suburban	1200 ft	7.0
C Open Terrain	900 ft	9.5
D Open Coast	700 ft	11.5

Averaging Time Of Wind

Surface winds in the atmospheric boundary layer are a turbulent flow, characterized by the random fluctuations of velocity and pressure. The wind speed used in structural design is the mean value averaged over a given time. The wind speed used in the United States prior to ASCE 7-95 was the fastest-mile wind; the peak wind speed averaged over 1 mile of wind passing through the anemometer. The averaging time of the fastest-mile wind is as follows:

$$T = \frac{3600}{V_F}$$

where:

T = Averaging time in seconds.

V_F = Fastest-mile wind in miles per hour.

The Canadian codes use an averaging time of 1 hour. ASCE 7-95 and current British and Australian codes use an averaging time of 3 seconds, the gust speed measured by ordinary anemometers. As the averaging time decreases, the mean wind speed for a given return period increases.

Because codes of different countries are based on different averaging times, their specified wind speeds cannot be compared without converting to the same basis. Similarly, empirically derived coefficients to be multiplied times wind speeds must be compared carefully to ensure their applicability.

Converting wind speeds from one averaging time to another can be done with the aid of ASCE 7-95, Figure C6-1. This figure converts all wind speed averaging times to an hourly average time. The scale factor is 1.00 at 3600 seconds. There are separate conversion scales for non-hurricane and hurricane winds. In the example below, a 70 mph "Fastest Mile" wind speed is converted to a 85 mph "3-Second Gust" wind speed.

$$T_{70 \text{ mph (fastest mile)}} = \frac{3600}{70} = 51 \text{ seconds}$$

$$70 \text{ mph (fastest mile)} \left(\frac{1.52}{1.26} \right) = 84.4 \text{ mph} \approx 85 \text{ mph (3-second gust)}$$

Where conversion factors 1.26 and 1.52 are obtained for non-hurricane winds from ASCE 7-95, Figure C6-1, for averaging times of 51 seconds and 3 seconds respectively.

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Gust Effect Factors

Wind gust is the instantaneous velocity of wind. Ordinary structures are sensitive to peak gusts of a duration of 1 second. It is customary to design structures to withstand gusts rather than the peak wind speed averaged over a longer time period. In general, the more flexible a structure, the more sensitive it is to gusts. Gust effect factors in the United States are based on the 3-second gust wind speed. Gust effect factors in Canada are based on the fastest hourly average. The two gust factors cannot be readily compared because of the different wind speed averaging times.

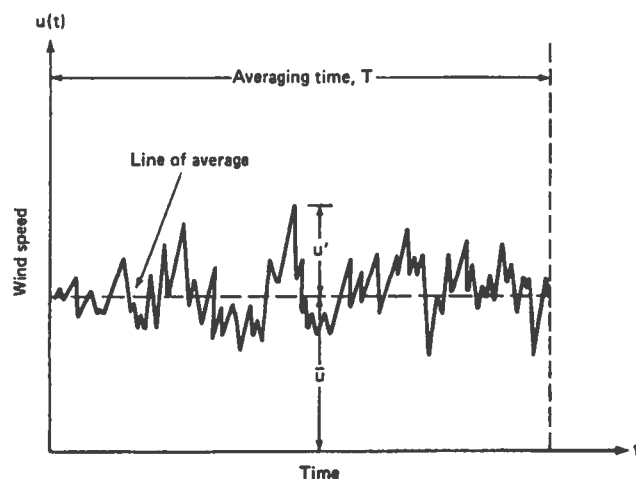


Figure 3.3 Typical trace of variation of wind speed u with time t .

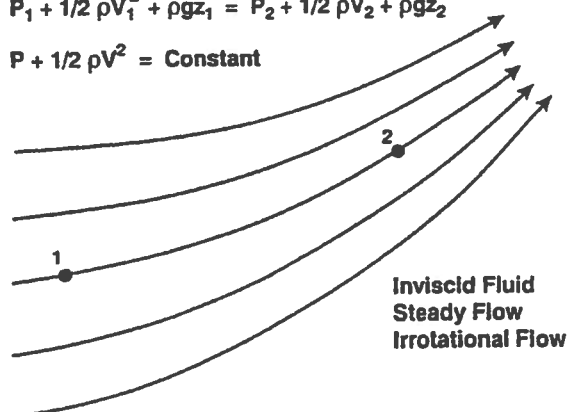
Bernoulli Effect

The equation that characterizes fluid flow is known as the Bernoulli Theorem. It compasses the essential balance between kinetic energy and potential energy over every part of a streamline in steady fluid flow.

BERNOULLI'S EQUATION

$$P_1 + 1/2 \rho V_1^2 + \rho g z_1 = P_2 + 1/2 \rho V_2^2 + \rho g z_2$$

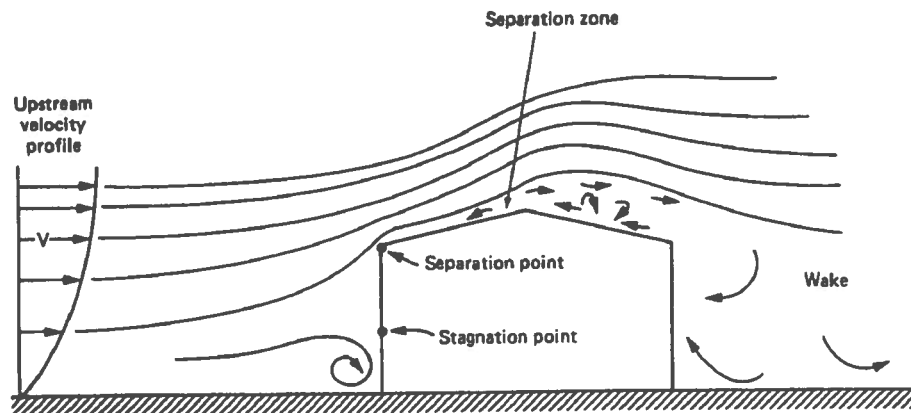
$$P + 1/2 \rho V^2 = \text{Constant}$$



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A steady fluid flow will increase in velocity when encountering an obstruction in its path. This increase in velocity will result in a decrease in pressure as demonstrated by the Bernoulli Theorem. This effect is responsible for the lift on an airplane wing and the suction pressures on the roof, side walls, and leeward wall of an enclosed structure. Further suction pressures are introduced at the sharp edges of structures where the fluid flow separates from the structure. These flow separation areas are called the wake region. Because the wake region is separated from the fluid flow, the Bernoulli Theorem cannot be used, and pressures are empirically determined in a wind tunnel.



WIND PRESSURES AND FORCES ON STRUCTURES

The pressure on the surface of a structure is the force per surface area exerted perpendicular to the surface. The reference pressure is the ambient pressure before wind flow. A positive pressure (above ambient) acts toward the surface. A negative pressure (below ambient) is called a suction and acts away from the surface.

Stagnation Pressure

The only place at which the external pressure on a structure can be accurately predicted from theory is at the stagnation point, located slightly above the center of the windward surface.

Assuming that the velocity of wind at the stagnation point is 0.0, the Bernoulli Theorem yields the following result:

$$P_s = \frac{\rho V^2}{2}$$

where

P_s = Stagnation pressure (also known as dynamic pressure or velocity pressure).

Pressure Coefficients

The local pressures at points on a structure are conveniently expressed as functions of the stagnation pressure as follows:

$$C_p = \frac{P}{P_s}$$

where

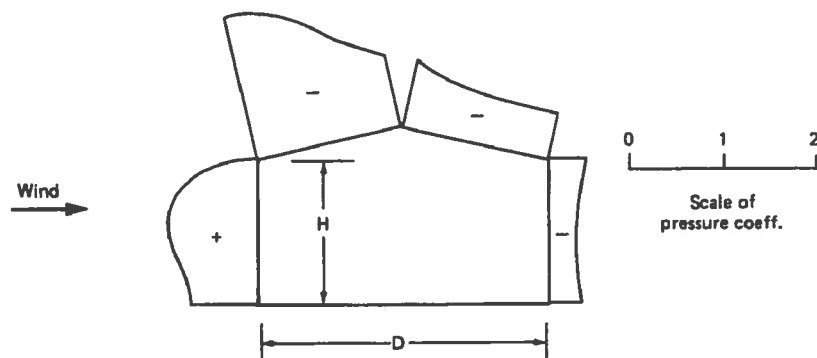
C_p = Pressure coefficient, calculated for different points on a structure.

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P = Pressure, determined empirically for different points on a structure.

Pressure coefficients are usually presented in dimensionless form. In dimensionless form, pressure coefficients are valid for almost any wind speed and air density, as long as the shape of the building and the orientation of the wind is fixed. This form allows pressure coefficients to be determined empirically in wind tunnels and to be applicable to the design of structures with the same shape.



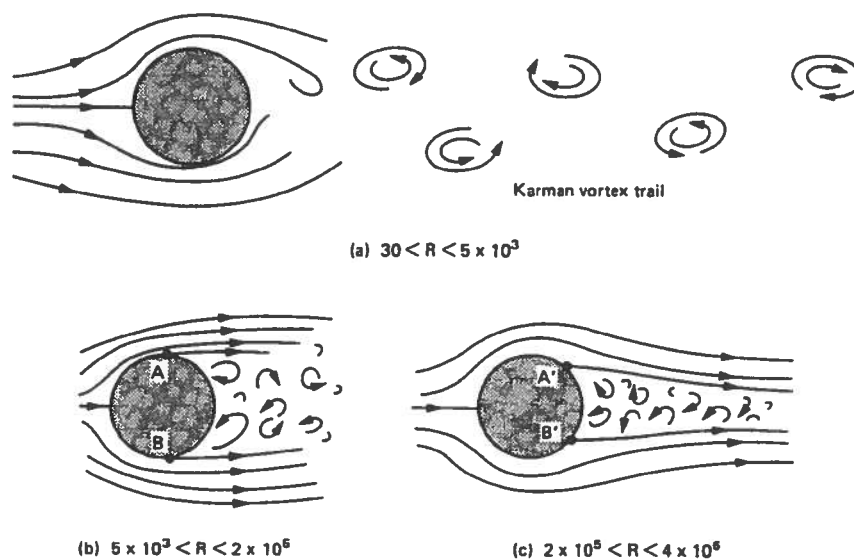
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ACROSS-WIND RESPONSE

PROCEDURES FOR VERTICAL VESSELS, STACKS, AND CHIMNEYS

Across-Wind Response

Wind flow past a circular cylinder can form vortices that shed from opposite sides of the cylinder at regular frequency. These alternating differential forces cause lift forces in the direction perpendicular to the direction of wind flow. The procedure for the evaluations of across-wind response follows that of ASME STS-1-1992. Additional discussion can be found in the references by Liu and McBean.



Critical Wind Speed

The critical wind speed of the vessel is determined from the following formula:

$$V_c = \frac{n_1 D}{S_t}$$

where

V_c	=	Critical wind speed	(feet per second)
n_1	=	First mode frequency	(Hertz)
D	=	Mean diameter of upper third of vessel	(feet)
S_t	=	Strouhal number, usually taken as 0.2 for single stacks and may vary due to Reynolds numbers and multiple stacks	(dimensionless)

Mean Hourly Design Wind Speed

The mean hourly design wind speed of the vessel is determined from the following formulae:

$$V_D = 0.96 V_3 \sqrt{K_z}$$

or

$$V_D = 1.18 V_f \sqrt{K_z}$$

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ACROSS-WIND RESPONSE

where:

V_D	=	Design wind speed	(feet per second)
V_3	=	Basic wind speed (3-second gust -- used with ASCE 7-95)	(miles per hour)
V_f	=	Basic wind speed (fastest mile -- used with ASCE 7-93 and earlier)	(miles per hour)
K_Z	=	Velocity pressure exposure coefficient determined at Z equals vessel height	(dimensionless)

Across-Wind Evaluation

The evaluation of a vessel for across-wind response requires the determination of the parameter M from the following formula:

$$M = \frac{m \beta}{\rho D^2} \quad (\text{dimensionless})$$

where

m	=	Average mass of upper third of vessel per unit length	(k-sec ² /ft ²)
β	=	Structural damping expressed as a fraction of critical damping	(dimensionless)
ρ	=	Mean mass density of air = 2.38×10^{-6}	(k-sec ² /ft ⁴)

Across-wind response evaluation considerations are tabulated in the table below.

ACROSS-WIND RESPONSE EVALUATION CONSIDERATIONS

	M < 0.4	0.4 < M < 0.8	M > 0.8
$V_c > 1.3V_D$	Across-wind response is not a concern		
$0.4V_D < V_c < 1.3V_D$	Large vessel deflections (0.4D to 1.0D) are probable and measures must be taken to reduce the motion.	Large vessel deflections (up to 0.4D) are possible. Magnitude of motion must be evaluated for acceptability with respect to fatigue and aesthetics.	Across-wind response may exceed along-wind drag forces V. Refer to ASME STS-1-1992 for further guidance.
$0.2V_D < V_c < 0.4V_D$			Across-wind response is not significant for the fundamental frequency.
$V_c < 0.2V_D$			The second mode frequency should be checked.

Reduction Of Across-Wind Vibrations

Vibrations in tall slender vessels due to across-wind response can be reduced either by modifying the aerodynamic load or by modifying the vessel dynamic properties. It is not usually practicable to modify the vessel height and diameter because these are usually determined to meet process requirements. The following remedial measures may be appropriate:

- Increase the vessel stiffness.
- Increase the vessel mass.
- Increase the vessel damping.

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ACROSS-WIND RESPONSE

- Add vortex spoilers to the vessel. ASME STS-1-1992, Section 5.4, discusses strakes and shrouds and recommends dimensions for them.